Read, connect, explain, practice

Source slides 1-3 / An original neutral example for any course or subject.

The question behind this lesson

How can a source become a connected explanation and then a useful practice task? Read the whole source, group its consecutive ideas, explain their links, and try a task that uses those ideas. A separate source log records what you checked.

What our counts mean

The example log is invented for this demonstration. Each item is listed once and is either checked against the source or pending. A checked item can still be difficult to understand. These counts measure the log's coverage, not mastery or learning quality.

Meet the symbols before calculating

Symbol	Quantity	Meaning	Unit
N	Listed items	All distinct items in the chosen source log.	count
C	Checked items	Listed items checked against the source.	count
U	Unchecked items	Listed items still awaiting a source check.	count
q	Checked fraction	Checked items as a share of listed items.	no unit

N is the total; C and U are its non-overlapping parts. The checked fraction q compares C with N. Homework begins a new log with new counts.

Follow the source's logical route

Slide 1: connect the study actions. **Slide 2:** understand the count and its fraction. **Slide 3:** identify pending work and choose a next action. Pages 6-7 apply the same reasoning to a complete practice question.

Keep knowledge blocks connected

Source slide 1 / Read the whole argument before explaining one step.

1. Read: discover the source's question

Look through all source pages first. Identify the main question, the starting assumptions and the result. Keep page numbers visible so the explanation can be traced back to its source.

2. Connect: group consecutive ideas

A knowledge block may span several neighbouring pages. Group them because they answer one question, not because each contains one formula. Keep the original order and say what one block establishes before the next block begins.

A transition to say out loud

We have identified the whole set of listed items. Next we need to know which of those items were checked. That distinction will let us calculate a fraction, and the remaining items will guide the next source check.

3. Explain: show the missing steps

Define unfamiliar concepts before using them. For a calculation, introduce every symbol beside its first formula, show one transformation per line and state its reason. For an argument, connect a claim to its evidence. For code, show how each step changes the state.

4. Practice: reuse the connection

Include the entire practice question, its conditions and its subparts. Point to the blocks that supply each prerequisite. Then solve and check the task, explaining why the method applies. The source log helps us locate unfinished checking; an attempted task gives evidence about reasoning.

Next: slide 2 compresses a fraction into one line. Page 3 expands its meaning and calculates it without turning the number into a claim about understanding.

Understand the fraction first

Source slide 2 / The denominator is the whole; the numerator is the chosen part.

1. State the definition and its condition

N means distinct listed items; C means items checked against the source; q means the checked fraction. We require N greater than zero and C between zero and N. Dividing by N asks how much of the whole list belongs to the checked group.

$$q = \frac{C}{N}$$

Definition: put the checked count above the fraction bar and the total count below it.

$$q = \frac{6}{8}$$

Substitute: this invented log has C = 6 and N = 8.

$$q = \frac{3}{4}$$

Simplify: divide both numerator and denominator by 2. The share stays the same.

$$q = 0.75$$

Divide: three divided by four is 0.75, a number between zero and one.

$$100\,q = 75$$

Convert to per hundred: multiply the share by 100.

Interpret the percentage

The checked fraction is 75%: 6 of the 8 listed items were checked. A percentage is a share expressed per hundred; this does not imply there were 100 actual items. The percentage reports source-check coverage, not how well anyone understands the material.

Connect the checked and pending parts

Source slide 3 / Derive the link one justified transformation at a time.

N is the total, C is checked and U is pending. Every item belongs to exactly one group, so C + U = N and U = N - C. The new symbol r means the **pending fraction**: U divided by N. Both q and r refer to the same positive total N.

$$q + r = \frac{C}{N} + \frac{U}{N}$$

1. Use the definitions: q describes the checked group; r describes the pending group.

$$q + r = \frac{C + U}{N}$$

2. Add equal-denominator fractions: combine the counts and keep the same whole N.

$$q + r = \frac{C + (N - C)}{N}$$

3. Substitute U = N - C: pending items are the total minus checked items.

$$q + r = \frac{N}{N}$$

4. Cancel +C with -C: the numerator returns to the full count N.

$$q + r = 1$$

5. Divide N by itself: this works because N is positive.

$$r = 1 - q$$

6. Subtract q from both sides: the pending share is what remains of the whole.

Why this connection matters

For the first log, q is 0.75, so the pending share is 0.25, or 25%. This conclusion depends on two exclusive statuses covering the same complete list. It would fail if items were duplicated or if a third status were left out.

Choose a useful next study action

Interpretation bridge / Connect the record to explanation and practice.

1. Check the record before using the result

Confirm what counts as an item and use that definition consistently. Count each item once. Use the same list for N, C and U; do not compare a checked count from one source with a total from a different source. An empty list has no defined checked fraction.

What the result can and cannot establish

The 75% result tells us that 6 of 8 listed items were checked. It does not measure item difficulty, explanation quality or understanding. A difficult idea and a short definition both count as one item under this simple rule.

2. Continue the logic in different subjects

For an argument: explain how a claim follows from its evidence and identify an assumption.
For a procedure: explain why each step is needed and check the result against its conditions.
For code: trace a small input, describe the changing state and test a boundary case.
For a calculation: identify the model, define symbols, derive the relation and check the units.

3. Turn a gap into one concrete task

A pending item suggests a source check: locate the passage, verify what it says and then update the log. A checked but confusing item suggests an explanation task: state the question, show the missing step, and try a small example. These actions address different needs.

Connection to homework: the next pages use a new synthetic log. We keep the same definitions and show how the earlier fraction and complement relations answer each part.

Read the full question, build a plan

Integrated practice / The complete original question comes before its solution.

Original authored practice question

An original practice source log lists 10 distinct items. Seven items have been checked against that source; the other items are pending. No item is listed twice, and every listed item has exactly one of these two statuses. (a) Find the checked fraction and write it as a percentage. (b) Find the number and fraction of pending items. (c) Verify that the two fractions add to one. (d) Does the checked percentage prove that the learner understands the source? Explain your answer and propose one next study action.

1. Reset the example and identify what is given

This is a **new log**: N = 10 and C = 7. The counts from the first log no longer apply. U is the pending count, q the checked fraction and r the pending fraction. The question guarantees no duplicates and exactly two exclusive statuses.

2. Link each subpart to its prerequisite

(a) Page 3 defines the checked fraction and percentage.
(b)-(c) Page 4 explains the pending count, its fraction and the sum check.
(d) Page 5 separates what a count records from what an explanation demonstrates.

3. Solve part (a): start with the relation

$$q = \frac{C}{N}$$

Choose the definition: the question asks for checked items as a share of the whole list.

$$q = \frac{7}{10}$$

Substitute the new counts: use C = 7 and N = 10; do not reuse 6 and 8.

Continue on page 7: evaluate this fraction, find the pending group and interpret what the result means. Every subpart uses a relation already introduced in the lesson.

Complete and interpret the practice

Parts (a)-(d) / Calculate, check the whole and choose a useful next action.

4. Finish part (a), then find the pending group

$$q = 0.7$$

(a) Divide 7 by 10: 0.7 is the checked share, equivalent to 70%.

$$U = 10 - 7$$

(b) Subtract the checked count from the total: pending items are U = N - C.

$$U = 3$$

Evaluate: three listed items are still pending.

$$r = \frac{3}{10}$$

Use the pending-fraction definition: r = U divided by the same total N.

$$r = 0.3$$

Divide: the pending share is 0.3, equivalent to 30%.

$$q + r = 0.7 + 0.3$$

(c) Substitute both shares: the two groups cover the same whole list.

$$q + r = 1$$

Add: the shares total one, consistent with the assumptions.

Part (d): interpret and act

No. Checking 70% of listed items does not prove understanding. Choose one checked item, explain how it connects to its preceding idea, then attempt a task using that connection. Alternatively, verify one pending item against the source and update the log.

Try a fresh question and explain it

Retrieval practice / Use the same reasoning with new records, then state its limits.

Try before reading the answer

Two new synthetic logs use the same item definition. Log A has 5 listed items and 4 checked; log B has 10 listed items and 6 checked. (a) Which has more checked items? (b) Which has a larger checked fraction? (c) Find each pending fraction. (d) Does the larger fraction establish better understanding?

Separate a count from a fraction

(a) Log B has more checked items: 6 is greater than 4. **(b)** Compare fractions because the totals differ. Here q_A and q_B are the checked fractions of logs A and B; the subscripts label the log.

$$q_A = \frac{4}{5}$$

Substitute log A: its checked count is 4 and its total is 5.

$$q_A = 0.8$$

Divide: A has an 80% checked share.

$$q_B = \frac{6}{10}$$

Substitute log B: its checked count is 6 and its total is 10.

$$q_B = 0.6$$

Divide: B has a 60% checked share. A's fraction is larger, despite its smaller count.

Reuse the complement and explain the limit

(c) Page 4 gives r = 1 - q. A's pending share is 0.2 (20%); B's is 0.4 (40%). **(d)** Neither fraction establishes understanding. These are records of source checks, not scores for reasoning. To examine reasoning, explain an idea and attempt a suitable task.

Final check: the method needs a positive total, a consistent item definition, no duplicate items and two statuses covering the whole. If those conditions change, revisit the definition before reusing a formula.